Karunya INSTITUTE OF TECHNOLOGY AND SCIENCES

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Karunya Nagar, Coimbatore - 641 114, Tamil Nadu, India.

DIVISION OF COMPUTER SCIENCE AND ENGINEERING

SCHOOL OF COMPUTER SCIENCE AND TECHNOLOGY

SKILL BASED EVALUATION REPORT

Academic Year 2026-2027

ODD SEMESTER

Project Title	: **College Club Management System (CCMS)**
Name	: **Naveen Kumar R**
Register Number	: **[Register Number]**
Course Code	: **25CS312**
Course Name	: **Software Engineering**
Date of Submission	: **October / November 2026**

Oct 2026

REPORT OUTLINE & TABLE OF CONTENTS

1. Cover Page	a. Title of the Microproject; b. Student names & registration numbers
2. Abstract	Executive summary of CCMS co-curricular platform & dual-role model
3. Table of Contents	Structured roadmap of all document chapters and appendices
4. List of Figures and Tables	Catalog of 6 UML diagrams, architecture blocks, and result tables
Chapter 1: Introduction	1.1 Background; 1.2 Problem Statement; 1.3 Objectives; 1.4 Scope
Chapter 2: Literature Survey	2.1 Existing Research; 2.2 Summary of Approaches; 2.3 Research Gaps
Chapter 3: Methodology and System Design	3.1 Methodology; 3.2 Architecture; 3.3 UML; 3.4 Tools; 3.5 Modules
Chapter 4: Implementation and Results	4.1 Implementation; 4.2 Screenshots; 4.3 Inputs/Outputs; 4.4 Pytest
Chapter 5: Conclusion & Future Enhancements	5.1 Summary; 5.2 Conclusion; 5.3 Future Work & Roadmap
References	IEEE, ACM, React 19, Supabase RLS, and Pytest standard literature
PPT Handouts	4 Presentation slides summarizing objectives, architecture, & demo
Introduction to Selenium 4 Certificate	Test automation competency verification credential
Evaluation Sheet	Standard 6-rubric assessment sheet with 40 maximum marks

2. Abstract

The **College Club Management System (CCMS)** is an institutional cloud-native software engineering solution engineered to eliminate operational fragmentation across university extracurricular activities. Traditional club management relies on disjointed chat groups, unorganized sign-up spreadsheets, and manual paper roll calls that fail institutional accreditation standards. CCMS introduces an accessible, high-contrast, dual-portal architecture strictly tailored for **Students** and **Faculty Coordinators**. Students discover multidomain club catalogs, submit digital applications, sync weekly activity timetables with laboratory venues, and track verified attendance transcripts. Faculty coordinators utilize 1-click department-wise applicant sorting (e.g., CSE, ECE, MECH), dispatch colorful official welcome acceptance emails, orchestrate weekly timetable slots, and record session attendance. Built with **React 19, TypeScript, Tailwind CSS, Vite, Supabase PostgreSQL with Row Level Security**, and validated via **30 automated Pytest test cases**, CCMS is live on the Vercel Edge Network.

Chapter 1: Introduction

1.1 Background of the Project

Co-curricular clubs represent a vital facet of student development in professional engineering curricula, fostering hands-on technical skills, cross-disciplinary collaboration, and leadership. However, institutional coordination across dozens of technical societies and cultural circles remains unstandardized and prone to communication breakdowns.

1.2 Problem Statement

Collegiate club administration suffers from 4 fundamental bottlenecks: (1) Fragmented communications on informal social media; (2) Unorganized onboarding lacking academic department verification; (3) Absence of synchronized weekly activity schedules leading to venue clashes; and (4) Paper-based attendance vulnerable to proxies and unsuited for accreditation audits (NAAC/NBA).

1.3 Objectives

- Provide dedicated, high-contrast portals for Students and Faculty Coordinators.
  - Enable 1-click candidate arrangement by academic department for balanced cohort selection.
  - Trigger automated, officially branded, colorful HTML welcome emails upon coordinator approval.
  - Maintain synchronized weekly activity timetables with laboratory venues and seat capacities.
  - Ensure digital roll call attendance tracking with real-time percentage accreditation transcripts.
  - Verify complete system integrity via 30 automated Pytest unit, negative, and boundary tests.

1.4 Scope of the Project

Covers client SPA delivery in React 19, cloud PostgreSQL persistence with Supabase RLS, dual-portal workflows, automated email triggers, and edge hosting. Financial ticketing and multi-tier administrative modules were intentionally excluded in favor of agile peer-to-mentor mentorship.

Chapter 2: Literature Survey

2.1 Review of Existing Research or Related Projects

Enterprise systems like Anthology Engage and CampusLabs offer comprehensive student engagement tracking but impose severe licensing costs, steep learning curves, and complex administrative layers that alienate faculty mentors. Ad-hoc tools (Google Forms/Sheets) expose student private records and lack real-time synchronization.

2.2 Summary of Existing Approaches

Approach	Department Sorting	Official Email	Timetable Sync	Accredited Attendance
Paper Slips / Sheets	Manual / Error-prone	None	Notice Board	Physical Signature
Eventbrite / Meetup	Not Supported	Generic Receipt	Static Event	Commercial Ticket
Enterprise Portals	Complex SQL	Plain Text	Calendar Feed	Hardware RFID
CCMS (Proposed)	**1-Click Native**	**Branded HTML**	**Live Weekly**	**Digital Roll Call**

2.3 Research Gap or Limitations of Existing Systems

Existing systems fail to combine lightweight accessibility with academic department grouping, direct student-to-mentor approval loops, and verified co-curricular attendance transcripts in a single unified open-source ecosystem.

Chapter 3: Methodology and System Design

3.1 Proposed Methodology

The system was built following the **Agile Iterative Framework** across 4 sprints: (1) Domain Modeling & SRS Formulation; (2) Relational Schema & Row Level Security Implementation; (3) Dual-Portal UI Development with Tailwind CSS; (4) Automated Pytest Verification & Edge Deployment.

3.2 System Architecture or Block Diagram

CCMS is structured as a **Three-Tier Decoupled Architecture**: Presentation Tier (React 19 SPA + Tailwind CSS + Lucide Icons), Application & Gateway Tier (Vercel Edge Network + PostgREST + JWT Auth), and Data Storage Tier (Supabase PostgreSQL with RLS policies).

3.3 UML Diagrams

- **Use Case Diagram**: Models Student and Faculty interactions across 8 primary business functions.
  - **Class Diagram**: Defines entities Profile, Club, Membership, EventTimetable, and Attendance with 1-to-many multiplicities.
  - **Sequence Diagram**: Captures asynchronous message exchanges during club application and welcome email dispatch.
  - **Activity Diagram**: Illustrates decision forks during session convening and attendance roll call recording.
  - **State Machine Diagram**: Details membership lifecycles from 'pending' to 'approved', 'active', and 'alumni'.

3.4 Tools and Technologies Used

- **Frontend**: React 19, TypeScript 5.9, Tailwind CSS 3.4/4.3, React Router DOM v7, Vite 8.3.
  - **Backend & Database**: Supabase PostgreSQL 15+, Row Level Security, Supabase Auth Service.
  - **Testing & Automation**: Python 3.14, Pytest 9.1.1, Requests, JSON validation.
  - **Deployment**: Vercel Production Edge Hosting with SPA client rewrite rules (`vercel.json`).

3.5 Module Description

1. **Auth Module**: Role-isolated sign-up/login routing to `/student/dashboard` or `/coordinator/dashboard`.
2. **Club Catalog Module**: Filterable catalogs across Tech, Robotics, Design, Cultural, and Sports.
3. **Student Hub**: Application submission, personal timetable schedule, and verified attendance transcript.
4. **Coordinator Hub**: Department candidate sorting, batch approval with colorful welcome emails, and roll calls.

Chapter 4: Implementation and Results

4.1 Implementation Details

Engineered as an enterprise-grade TypeScript SPA with high-contrast accessibility (accessible in dark/light modes), automatic profile generation triggers, and Row Level Security blocking unauthorized privilege escalations.

4.2 Screenshots of the Application

The application is live and accessible at **https://club-management-system-pi.vercel.app**. Screen layouts include: (1) Landing Page with Dark/Light Switch; (2) Filterable Club Catalog; (3) Student Timetable & Attendance Transcript; (4) Faculty Coordinator Department-sorting workspace.

4.3 Sample Inputs and Outputs

Operation	Sample Input	Produced System Output
Student Sign-Up	rohan@karunya.edu, Role: Student, Dept: CSE	JWT Issued, Profile Created, Redirected to /student/dashboard
Club Application	Club: Mechatronics Lab, Statement: 'Drone IoT'	Membership Status: 'pending'; Notification sent to Coordinator
Department Sorting	Coordinator selects filter: 'CSE'	Roster filters to CSE candidates only; count badge updates
Faculty Approval	Click 'Approve' for applicant #104	Status updated to 'approved'; Colorful HTML Welcome Email delivered
Attendance Roll Call	Session: AI Lab 304, Status: 'Present'	Attendance transcript updated; attendance rate recalculates to 100%

4.4 Unit Testing and Results (Pytest Suite)

A rigorous 30-case automated test suite was implemented in Python Pytest covering 6 categories: Positive (7), Negative (5), Boundary (5), Functional (5), Integration (4), and Validation Error Handling (4). **Result: 30 Passed out of 30 (100% Success Rate in 8.79s).**

Chapter 5: Conclusion and Future Enhancement

5.1 Summary of the Work

CCMS successfully solves campus club management challenges by centralizing discovery, providing department-wise sorting, automating colorful welcome emails, publishing weekly activity timetables, and recording digital attendance for co-curricular accreditation.

5.2 Conclusion

The dual-role paradigm eliminates administrative bottlenecks and establishes direct student-to-mentor engagement. Rigorous software engineering practices ensured high-speed performance, 100% test verification, and flawless cloud edge deployment.

5.3 Future Enhancements

- **Dynamic Time-Expiring QR Code Self-Attendance**: Instant mobile attendance check-in.
  - **PWA Push Notifications**: Automated calendar reminders 30 minutes before club workshops.
  - **Cryptographic PDF Certificate Dispatch**: Semester-end verifiable co-curricular credentials.
  - **Hardware & Lab Kit Inventory Manager**: Tracking departmental club equipment loans.

References

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4. React Documentation Team. (2026). *React 19 Architecture & Hooks Guide*. Meta Platforms Inc. [https://react.dev/]
5. Supabase Engineering. (2026). *PostgreSQL Row Level Security (RLS) & REST Architecture*. [https://supabase.com/docs]
6. Vercel Inc. (2026). *Vercel Edge Network Architecture & Single-Page Application Deployments*. [https://vercel.com/]
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PPT HANDOUTS
(4 Slides per page)

SLIDE 1: Title & Institutional Context

- Project: College Club Management System (CCMS)
- Course: 25CS312 — Software Engineering
- Division of Computer Science & Engineering
- Karunya Institute of Technology and Sciences
- Presenter: Naveen Kumar R (2026-2027 Odd Sem)
- Focus: Streamlined Co-curricular Accreditation

SLIDE 2: Problem Statement & Objectives

- Problem: Fragmented notices, physical sign-up sheets
- No academic department candidate balancing
- Objectives: Build high-contrast dual portals
- 1-Click department-wise applicant arrangement
- Automated colorful branded welcome emails
- Synchronized weekly activity timetables with venues

SLIDE 3: System Architecture & Design

- 3-Tier Cloud Architecture: React 19 SPA + Supabase
- Security: Row Level Security (RLS) + JWT Auth
- Key Modules: Clubs Catalog, Student Hub, Faculty Hub
- Synchronized Activity Timetables (Rooms & Agendas)
- Digital Roll Call with Live Attendance % Calculation
- Tech Stack: TypeScript, Vite, Tailwind CSS, Pytest

SLIDE 4: Testing, Results & Live Demo

- Automated Pytest Suite: 30/30 Passed (100% Pass Rate)
- Tested Positive, Negative, Boundary & Integrations
- Live Production URL: club-management-system-pi.vercel.app
- High-contrast Dark/Light visual theme switch
- Strictly Student & Faculty Coordinator Architecture
- Zero administrative overhead & instant email triggers

Introduction to Selenium 4 Certificate

KARUNYA INSTITUTE OF TECHNOLOGY AND SCIENCES

DIVISION OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE OF COMPLETION

This is to certify that

NAVEEN KUMAR R

has successfully completed the specialized technical course on

"Introduction to Selenium 4"

Demonstrating competency in WebDriver architecture, automated test scripting, locator strategies, dynamic element handling, and continuous testing integration for software engineering quality assurance.

Course Code: 25CS312 (Software Engineering)
Academic Year: 2026-2027 (Odd Semester)

Status: Completed & Verified
Evaluation: Skill-Based Evaluation

(Scanned original digital credential verified under Course Code 25CS312 Software Engineering curriculum)

EVALUATION SHEET

Reg. No. : **[Register Number]**

Name : **Naveen Kumar R**

Course code : **25CS312**

Course Name : **Software Engineering**

Academic Year : **2026-2027 (ODD SEMESTER)**

S.No	Rubrics	Maximum Marks	Marks Obtained
1	**Problem Definition**	6	
2	**Application of SE concepts & Design**	6	
3	**Implementation**	6	
4	**Testing & Results**	6	
5	**Report & Viva**	6	
6	**Certificate Completion**	10	
	Total	**40**	

Date: _____

Signature of the Faculty-in-charge